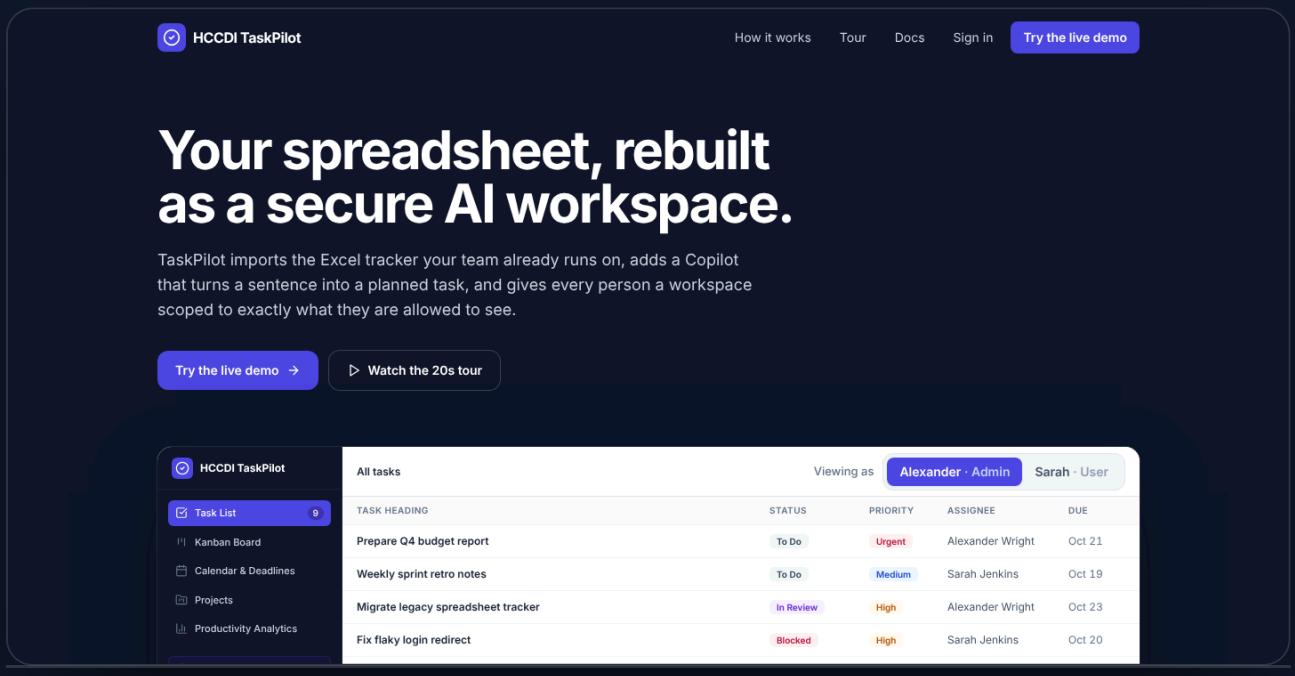
HCCDI TaskPilot

Product Proposal: your spreadsheet, rebuilt as a secure AI workspace.

A proposal to run HCCDI TaskPilot as the team's task system of record, replacing the shared Excel tracker, piloted with one team on Vercel and Neon PostgreSQL.



PREPARED BY
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VERSION
1.0 · Proposed

DATE
October 2026

LIVE DEMO
ai-based-to-do-program.vercel.app

1 Summary

Teams that coordinate work in a shared Excel workbook pay for it every week. Statuses drift ("Pending", "Under Review?", blank), nobody owns half the rows, every person can read every row, and there is no history of who changed what.

HCCDI TaskPilot replaces that workbook with a secure, AI-assisted task workspace, **without asking the team to abandon Excel on day one**. The existing sheet is imported with a validation preview, and a styled spreadsheet can be exported back at any time.

This proposal asks for approval to run TaskPilot as the team's task system of record, hosted on Vercel with a managed PostgreSQL database (Neon).

What it is	Web app: React SPA, FastAPI API, PostgreSQL
Who it is for	Admins (team leads, ops managers) and users (individual contributors)
Core promise	Spreadsheet in, secure AI workspace out
Live demo	ai-based-to-do-program.vercel.app: landing page, 20-second video tour, demo accounts on `/login`
Source	github.com/Hamzaviour/AI-Based-To-Do-App

2 Problem

PAIN IN THE SHARED WORKBOOK	CONSEQUENCE
Everyone sees every row	Confidential or personal work is exposed to the whole team
Free-text statuses and priorities	Reports disagree; "Pending" and "To Do" are counted separately
Rows without owners or titles	Work silently falls through the cracks
No audit trail	Nobody can tell who changed a deadline or closed a task
No planning help	People decide what to do next by scrolling

3 Proposed solution

Every capability below is implemented, tested and running on the live demo.

01 Excel round-trip

Upload `.xlsx` and review a dry-run preview: legacy statuses are mapped and rows missing a title are held back. Commit, then export back to `.xlsx` with native dropdowns on status and priority.

02 Isolated workspaces

Every task query passes through one scope filter. Users see only their own, assigned or project tasks. A probe for anyone else's task returns **404**, not 403, so existence isn't leaked.

03 AI Task Copilot

Turns a sentence into a task with title, priority, due date and checklist. It also breaks down work, explains priorities and plans the day. It works fully offline, and an LLM key can be added for richer output.

04 Four ways to work

A dense list with bulk actions, a drag-and-drop Kanban board and a monthly calendar. A slide-over drawer holds the checklist, comments and activity log. `Ctrl/⌘ K` opens the command palette.

05 Governance

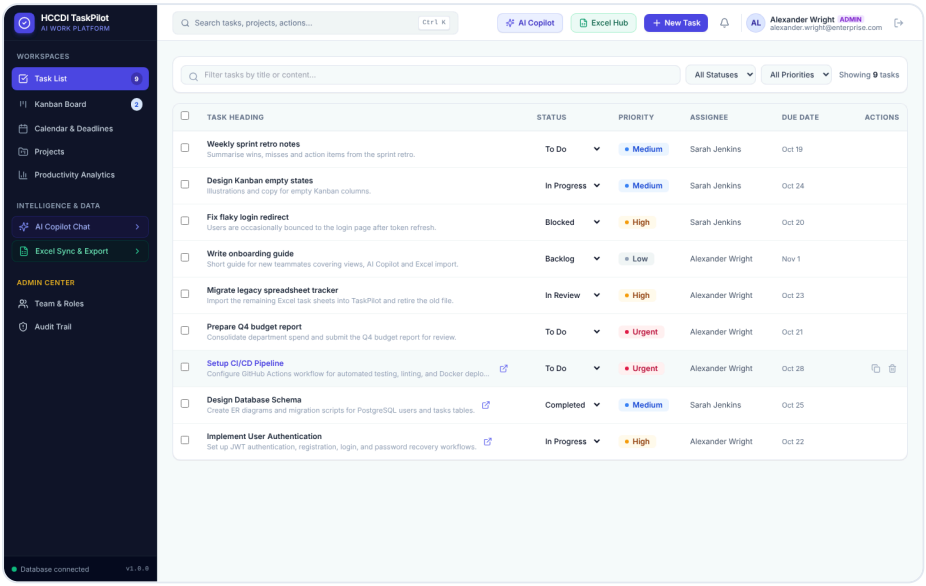
Admins provision users and assign roles, review the organisation-wide audit trail and see productivity analytics across the team.

Why this approach

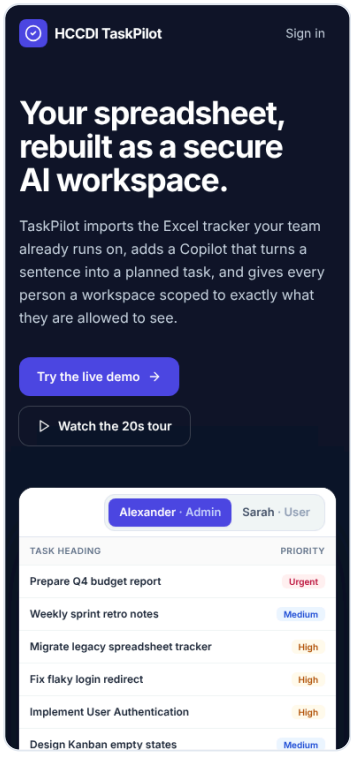
- **Low switching cost.** The import/export loop lets the team migrate gradually and keeps stakeholders who still want a spreadsheet.
- **Security by construction.** Isolation lives in the data-access layer (`get_scoped_tasks_query` / `get_task_or_404`), not in UI hiding, and is covered by automated tests.
- **No vendor lock on AI.** The Copilot runs without any API key; an LLM provider is optional.

The product today

Screens from the live deployment, signed in as the admin demo account.



Workspace: every task in the admin's organisation scope, with status, priority, owner and due date.



The landing page on a phone.

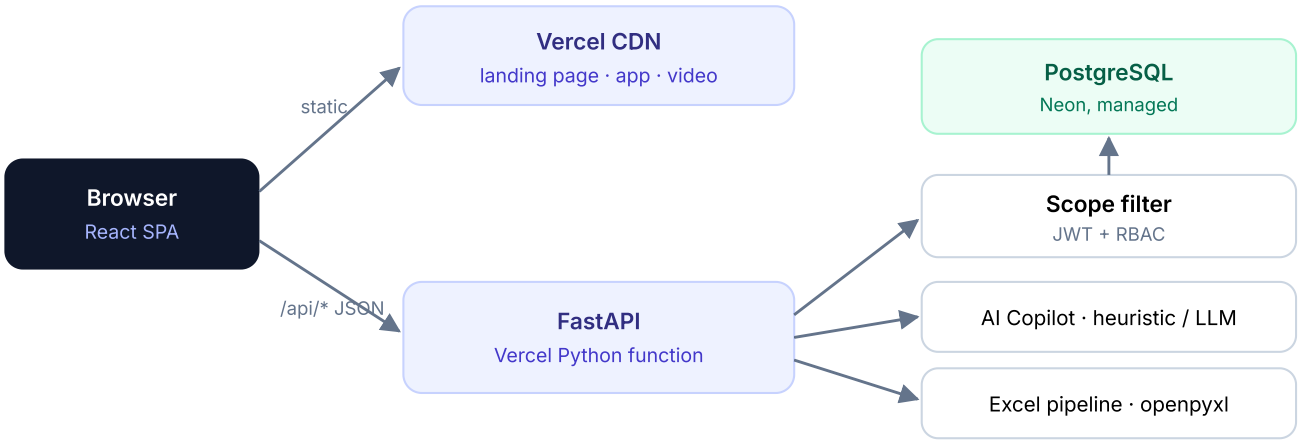
4 Scope

IN SCOPE FOR ROLLOUT	The capabilities above, demo data, the Vercel deployment and the documentation suite.
OUT OF SCOPE	Candidate follow-ups, not built: email/Slack notifications, SSO/SAML, a recurring-task automation UI, mobile apps and multi-organisation tenancy.

5 Users and roles

ROLE	CAN	CANNOT
Admin	See and edit every task; provision users and roles; read the audit trail and organisation analytics	—
User	Create tasks; work on tasks they own, are assigned, or share through a project	See other people's tasks (404); open admin pages (403)

6 Architecture at a glance



Every request from a user to their tasks crosses the scope filter before it reaches the database. The landing page, app and video are served from Vercel's CDN, and the API runs as a single serverless function.

7 Rollout plan

STEP	OUTCOME	OWNER	STATUS
1. Deploy	Vercel project and Neon database live, secrets set, demo data loaded	Engineering	Done
2. Pilot	One team imports its workbook via the preview and runs on TaskPilot for two sprints	Team lead	Next
3. Review	Compare against the success measures below; fix gaps	Team lead + Engineering	
4. Expand	Provision remaining users; retire the shared workbook (export kept as archive)	Admin	

8 Success measures

Proposed targets to agree with the sponsor, not measured results.

- 100% of active work items live in TaskPilot by the end of the pilot; the shared workbook becomes read-only.
- Zero cross-user data exposures, verified by the isolation test suite on every change.
- Every task has an owner and a valid status after import, enforced by the preview.
- Weekly stakeholder reports produced from the Excel export instead of hand-edited sheets.

9 Risks and mitigations

RISK	MITIGATION
Default secrets reach production	The API refuses to start on Vercel if `JWT_SECRET_KEY` is the public default
Messy legacy sheets	Dry-run preview; rows missing a title are held back and statuses normalised
AI quality without an LLM key	The deterministic engine is predictable; drafts are always editable and never saved automatically
Serverless cold starts	One lightweight function using Neon's connection pooler; static assets on the CDN
Adoption	Excel export stays available; the landing page and a 20-second video explain the product

10 Cost model

Hosting runs on two managed services, each with a free tier suitable for a pilot: **Vercel** (frontend and Python function) and **Neon** (PostgreSQL). Paid tiers are only needed for production-scale traffic or team features; check both vendors' current pricing before the expansion step. An LLM provider is optional and billed by usage if enabled.

11 Decision requested

Approve the pilot (steps 2 and 3) with one team on the live deployment.

Try it first: ai-based-to-do-program.vercel.app. Sign in as the admin and the isolated user, and compare what each can see.